Netflix Content Recommendation System

1. Objective

The objective of this task was to develop a basic content-based recommendation system that recommends Netflix titles similar to a selected title.

2. Methodology

Feature Extraction:
Relevant Netflix content features were combined into a `content` feature for comparison.

Text Preprocessing:
The text was converted to lowercase and processed using TF-IDF to transform textual information into numerical features.

Content Similarity:
Cosine similarity was used to measure the similarity between Netflix titles. The resulting similarity matrix contained **8,790 × 8,790** comparisons.

Recommendation Generation:
A recommendation function was created to identify the most similar titles and return the top five recommendations for a selected title.

3. Results

Examples of generated recommendations included:

- **Ganglands:** Sentinelle (0.759), Lupin (0.730)
- **Midnight Mass:** Brand New Cherry Flavor, The Haunting of Bly Manor, Ratched, and others (≈0.728)
- **Sankofa:** Residue (0.550)

4. Evaluation

The recommendation system was evaluated using **Precision@5**.

Title	Precision@5
Ganglands	60%
Midnight Mass	100%
Sankofa	100%
Dick Johnson Is Dead	0%

5. Findings

The model produced strong recommendations for some titles but performed differently depending on the selected title. This shows that recommendation quality depends on the features used to represent the content.

6. Conclusion

The task successfully demonstrated the use of **Python, Pandas, Scikit-learn, TF-IDF, cosine similarity, recommendation systems, and model evaluation** to build a basic Netflix content recommendation system.

The system could be improved by incorporating additional information such as descriptions, genres, cast, directors, and countries.